Functional Analysis

Latex Notes

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注意：本模板自 **2023** 年 **1** 月 **1** 日起停止维护。然而，鉴于用户群体庞大，自 **2026** 年起恢复维护并重新发布。

目录

第一章 Normed Spaces

定义 1.1

A V-S V is a **infinite-dimensional** if every linearly independent set is a finite set. I.e, for all sets $E \subseteq V$ s.t

$$\sum_{i=1}^{N} a_i v_i = 0 \quad \forall v_1, \dots v_N \in E,$$

E has finite cardinality. V is **finite-dimensional** if it is not infinite-dimensional. ♣

例题 1.1

$C([0,1])$ is infinite-dimensional. Consider the set $E = \{f_n(x) = x^n : n \in \mathbb{Z}_{\geq 0}\}$. This set is linearly independent and has infinite cardinality.

定义 1.2

Let X be a metric space. The vector space $C_\infty(X)$ is defined as

$$C_\infty(X) = \{f : X \to \mathbb{C} : f \text{ is continuous and bounded}\}$$

♣

It is obvious to see that $C_\infty([0,1])$ is $C([0,1])$, becuz all conts functions on [0,1] are bounded.

命题 1.1

For any metric space X, we define a norm on the vector space $C_\infty(X)$ as

$$\|u\|_\infty = \sup_{x \in X} |u(x)|$$

♠

Proof: First checking the norm. For triangle inequality, if $u, v \in C_\infty(X)$, then $\forall x \in X$, we have

$$|u(x) + v(x)| \leq |u(x)| + |v(x)|$$

Taking the supremum over $x \in X$, we get

$$\|u + v\|_\infty \leq \|u\|_\infty + \|v\|_\infty$$

and then we imposed supremum over both sides.

定义 1.3

The ℓ^p space is the space of (infinite) sequences

$$\ell^p = \{\{a_j\}_{j=1}^\infty : ||a||_p < \infty\}$$

where we define the ℓ^p norm

$$||a||_p = \begin{cases} (\sum_{j=1}^\infty |a_j|^p)^{1/p}, & 1 \leq p < \infty \\ \sup_{j \geq 1} |a_j|, & p = \infty. \end{cases}$$

♣

定义 1.4

A normed space is **Banach space** if it is complete w.r.t the metric induced by the norm. I.e, every Cauchy sequence converges in the space. ♣

例题 1.2 $\forall n \in \mathbb{Z}_{\geq 0}$, $\mathbb{R}^n$ and $\mathbb{C}^n$ are complete w.r.t any of the $|| \cdot ||_p$ norms.

定理 1.1

For any metric space X, the space of bounded continuous fnc on X is complete, thus $C_\infty(X)$ is a Banach space.

Proof:

定义 1.5

Let $\{v_n\}_{n=1}^{\infty}$ be a seqn of points in V. Then the series $\sum_n v_n$ is **summable** if $\{\sum_{n=1}^{m} v_n\}_{m=1}^{\infty}$ converges, and **absolutely summable** if it same statement hold with $||v_n||$.

命题 1.2

if $\sum_n v_n$ is absolutely summable, then the sequence of partial sum $\{\sum_{n=1}^{m} v_n\}_{m=1}^{\infty}$ is Cauchy.

定理 1.2

A normed vector space V is Banach i.f.f every abs summable series is summable.

Proof:

例题 1.3 Let $K : [0,1]x[0,1] \to \mathbb{C}$ be a continuous function. Then for any func $f \in C([0,1])$. we define

$$Tf(x) = \int_0^1 K(x,y)f(y)dy$$

The map T is basically the inverse operators of differentyial operators.

定义 1.6 (Linear Operator)

Let V & W be two V-S. A map T: V $\to$ W is **linear** if for all $\lambda_1, \lambda_2 \in \mathbb{K}$ and $v_1, v_2 \in V$, we have

$$T(\lambda_1 v_1 + \lambda_2 v_2) = \lambda_1 T(v_1) + \lambda_2 T(v_2)$$

$\boxed{Note}$:

All linear transformation are continous in finite-dimensional V-S. But this is not the case for infinite-dimensional V-S. Consider X to be n-dimensional V-S with basis $\{e_1, e_2, \ldots, e_n\}$. Let vector $x \in X$ be represented as a linear combination of the basis vectors,

Apply linearity:

$$T(x) = T\left(\sum_{i=1}^{n} x_i e_i\right) = \sum_{i=1}^{n} x_i T(e_i)$$

Applying Δ-inequality & taking the norm in Y:

$$||T(x)||_Y = ||\sum_{i=1}^{n} x_i T(e_i)||_Y \leq \sum_{i=1}^{n} |x_i| ||T(e_i)||_Y$$

Due to the equivalence of norms in finite-dimensional V-S, norm on X are equivalent with ℓ^1 norm, $\|x\|_1 = \sum_{i=1}^{n} |x_i|$ Thus we have

$$||T(x)||_Y \leq C\|x\|_X$$

for some constant C. This shows that T is continuous.

To bound the operator let $M = \max_{1\leq i\leq n} ||T(e_i)||_Y$. Then we have

$$||T(x)||_Y \leq \sum_{i=1}^{n} |x_i| ||T(e_i)||_Y \leq M \sum_{i=1}^{n} |x_i| = M\|x\|_1$$